Engineering portfolio

Thomas Lepère

M.Sc. in Mechanical Engineering - Fluid Mechanics specialization

CFD | Aerodynamics | Wind-tunnel testing | Data analysis | Correlation

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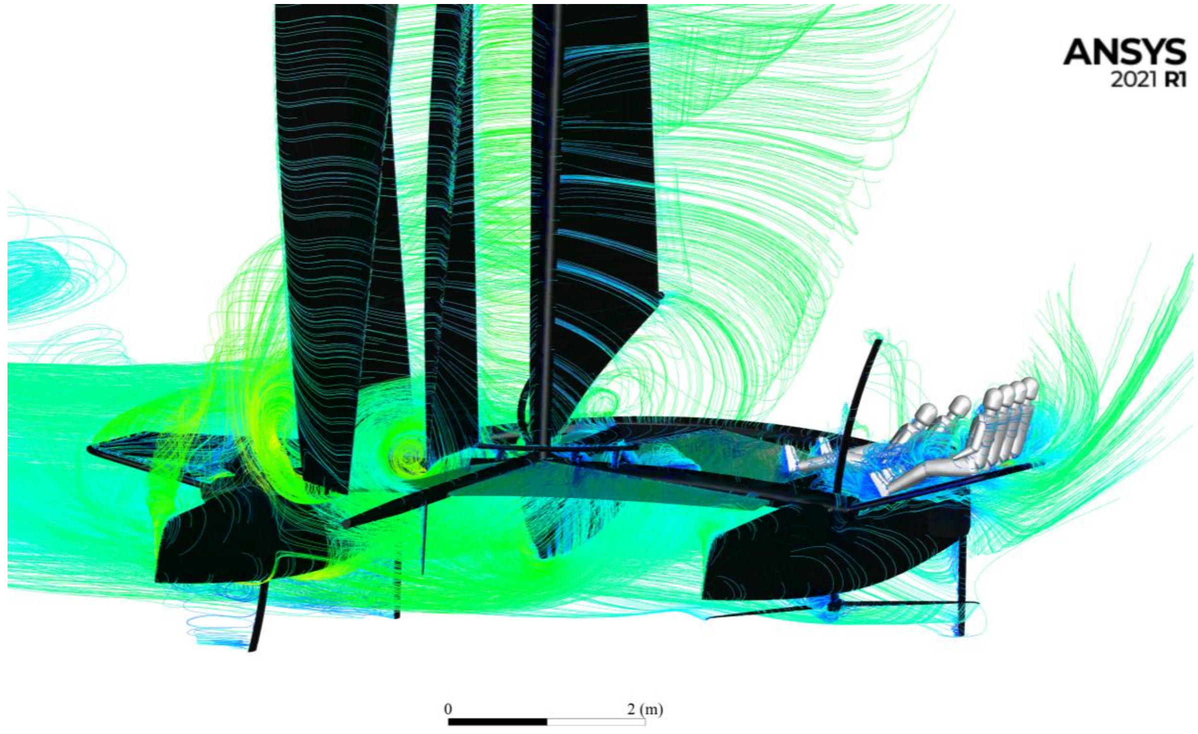
Selected projects

Built for internship applications – Concise project evidence, not a report

Formula student wind tunnel testing	Design / Wind tunnel testing
CFD correlation of a foiling boat	CFD validation methodology
Wind turbine data analysis	Data analysis + ML forecasting
CFD study of two-element wingsail	2D parametric RANS
Particle-laden buoyant jets	Environmental CFD
Impact of pneumatic tubes on pressure measurements	Experimental aerodynamics
Improving ski-helmet testing with a neck model	CAD + Experimental
Lift computation of a flapped 2D airfoil	Analytical aerodynamics
Crack-tip fields from Digital Image Correlation	Experimental + Nonlinear mechanics
Structural simulation of a quadcopter drone	Finite elements

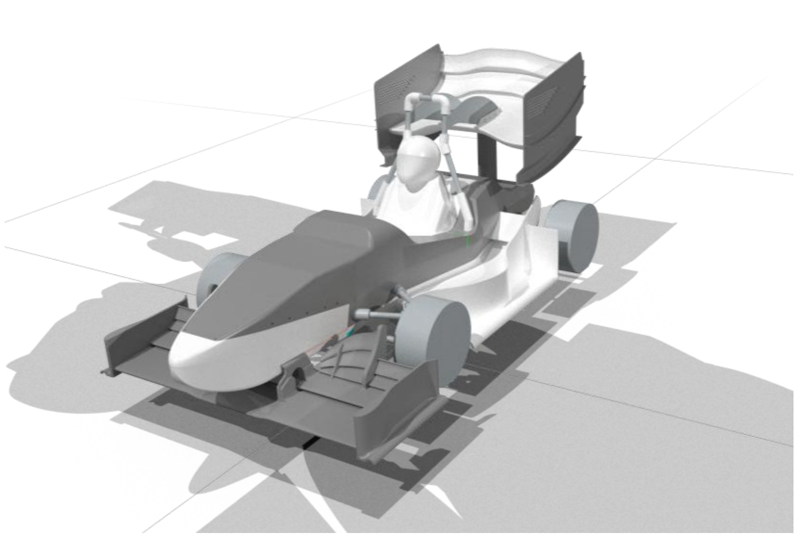


CFD/Wind-tunnel correlation for a Formula Student race car

Semester project – EPFL Racing Team

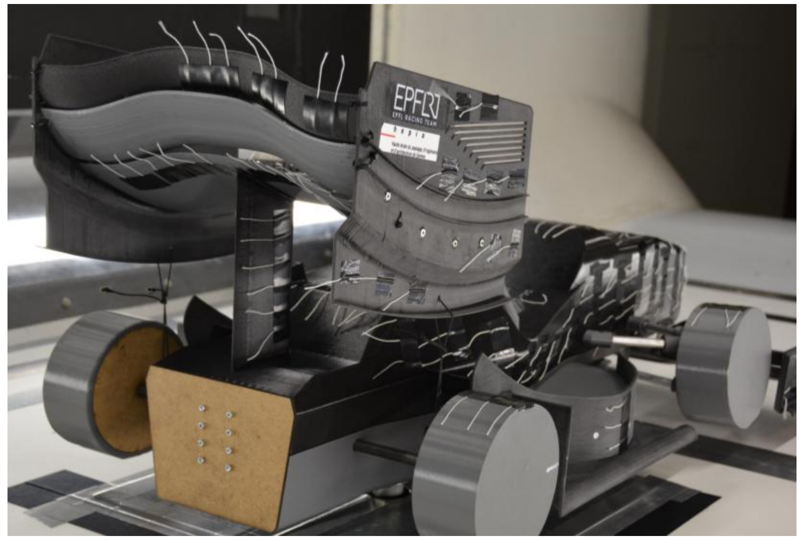
3DExperience, 3D printing, Python

Aerodynamics – Wind tunnel testing – CAD - Correlation



CAD of the reduced model

Photograph of real car's rear wing



Reduced model flow at 90 km/h

Reduced model in the wind tunnel

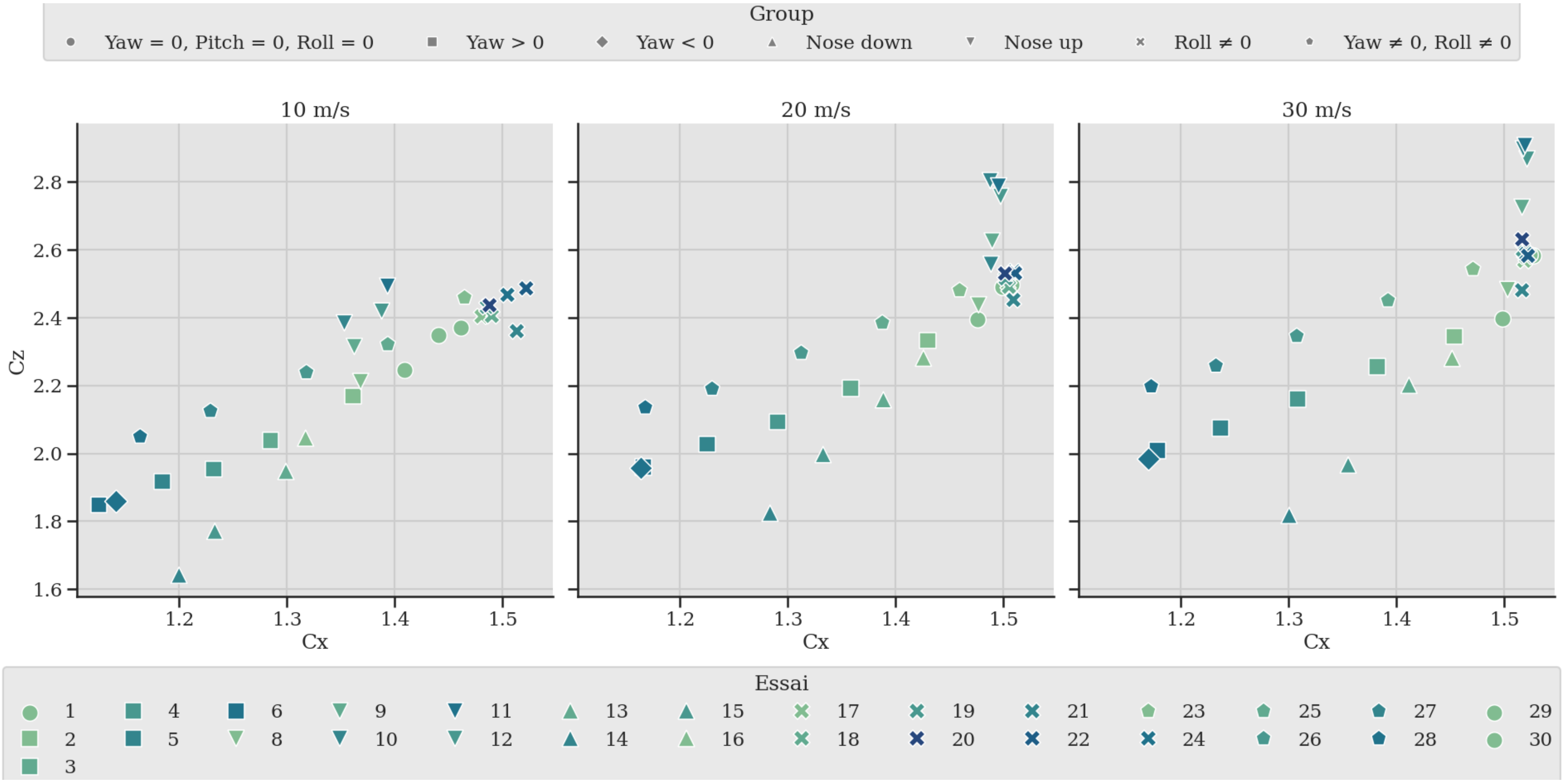
149 tests done

50 m/s maximum wind speed

Objective Validate the CFD methodology against reduced-scale wind-tunnel measurements.

Method Designed and tested a 1:3-scale 3D-printed model and compared measured aerodynamic coefficients with CFD predictions.

Results Validated drag coefficients within 10%; lift remained more sensitive to the experimental setup and measurement uncertainty.



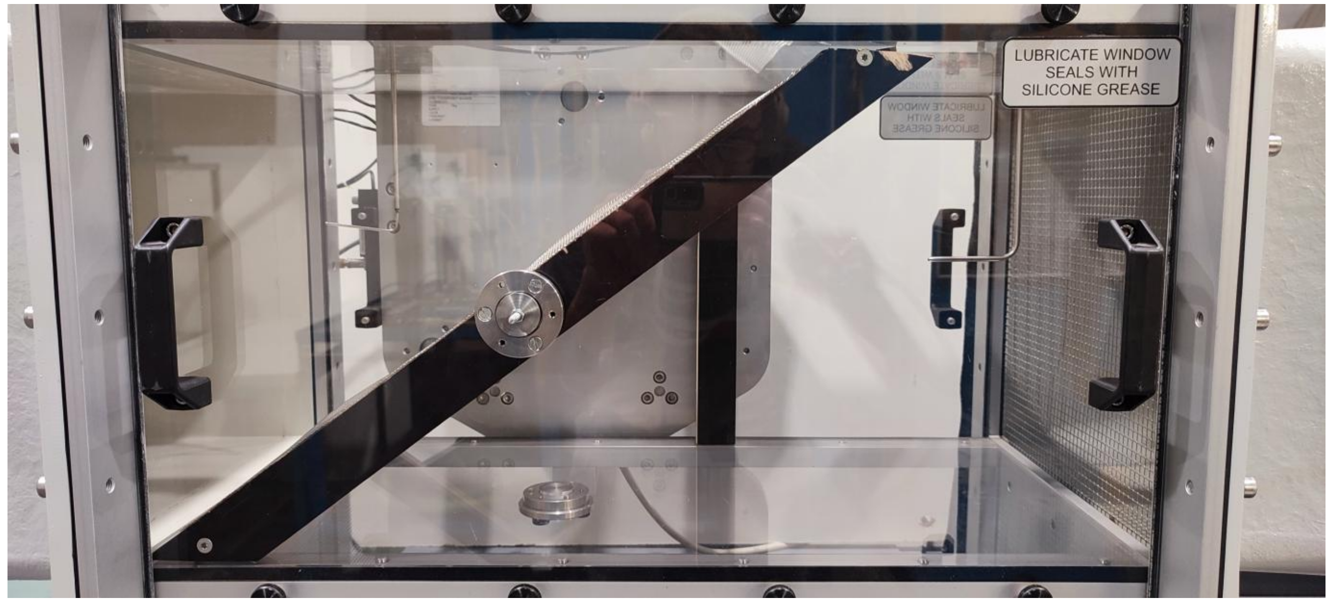
Results of the wind tunnel tests for 3 different wind speeds

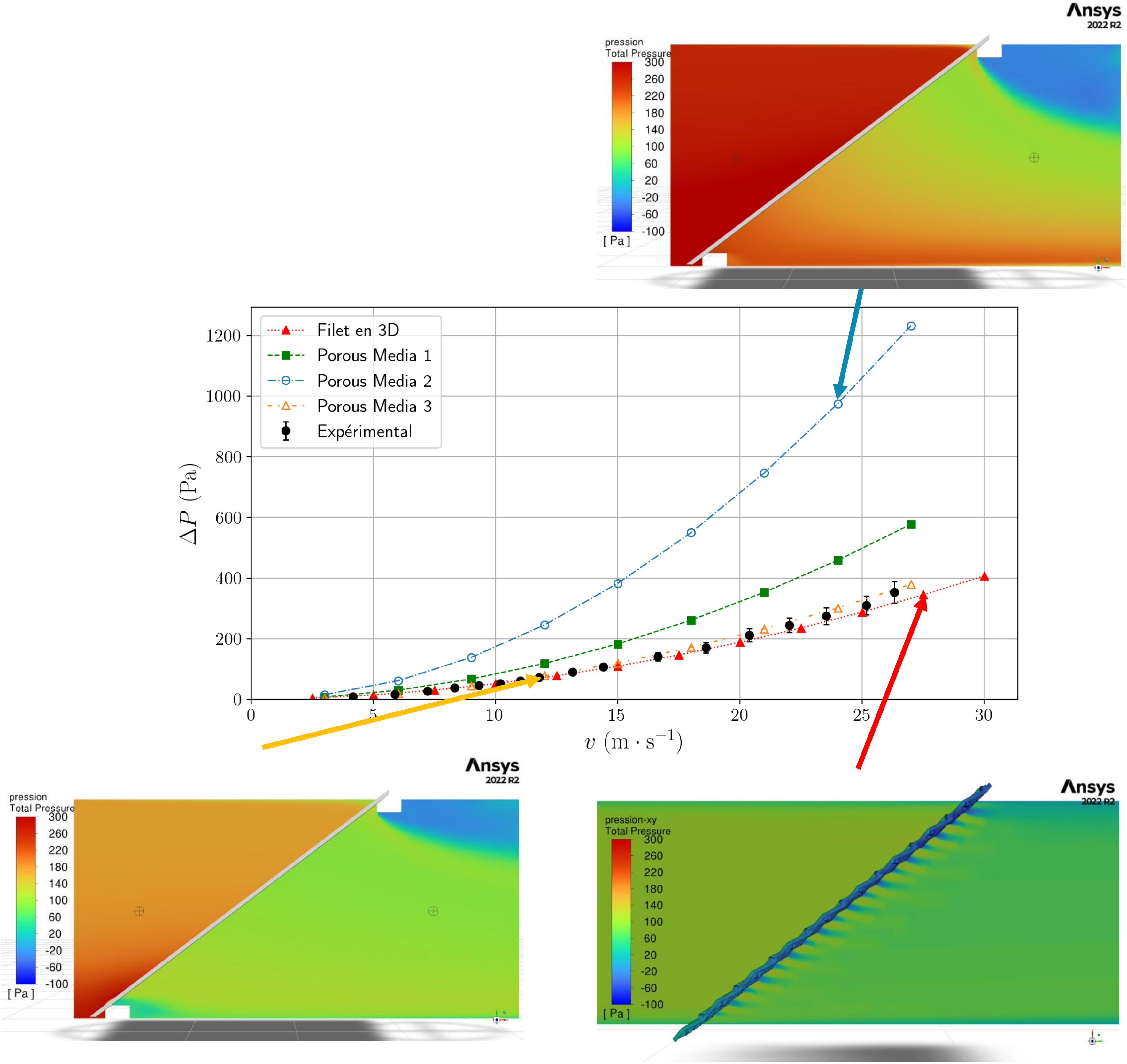
CFD/Wind-tunnel correlation for a foiling catamaran

Bachelor thesis - HEPIA

ANSYS Fluent, Python

Aerodynamics – Wind tunnel testing – CFD - Correlation





Objective Investigate the discrepancy between previous wind-tunnel measurements and CFD drag predictions.

Method Reproduced reduced-scale wind-tunnel conditions and compared fully resolved and porous-media models of the netting.

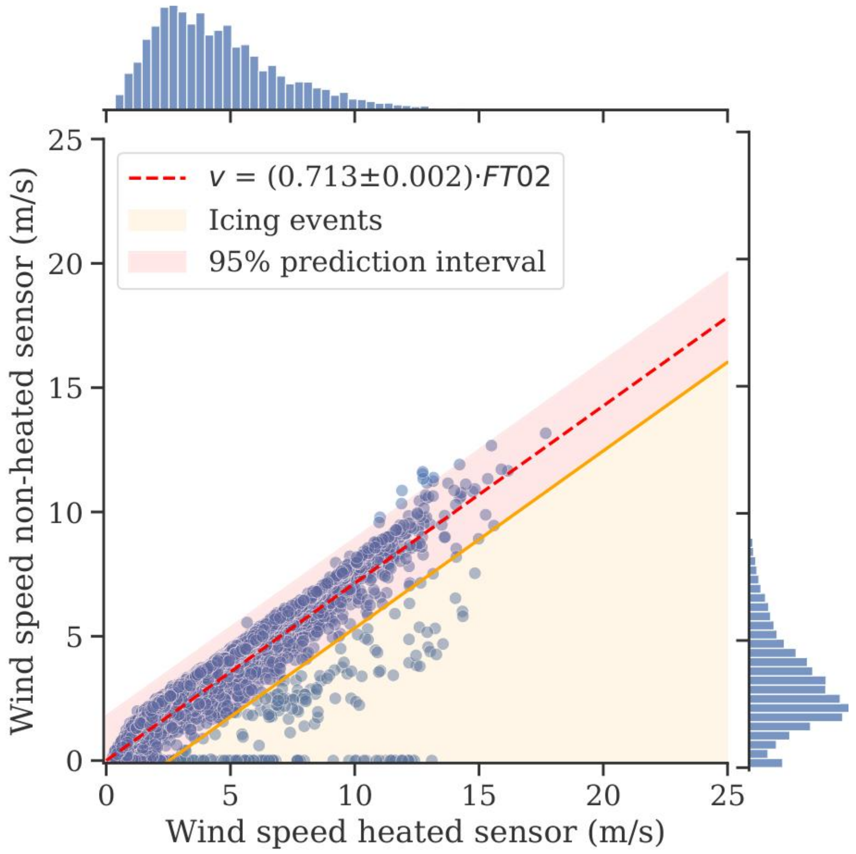
Result Matched oblique-incidence pressure losses within ±10% at component level, but showed that the porous-media approach alone could not reproduce the full net behaviour.

Estimating wind-turbine energy losses from icing and long-term wind predictions

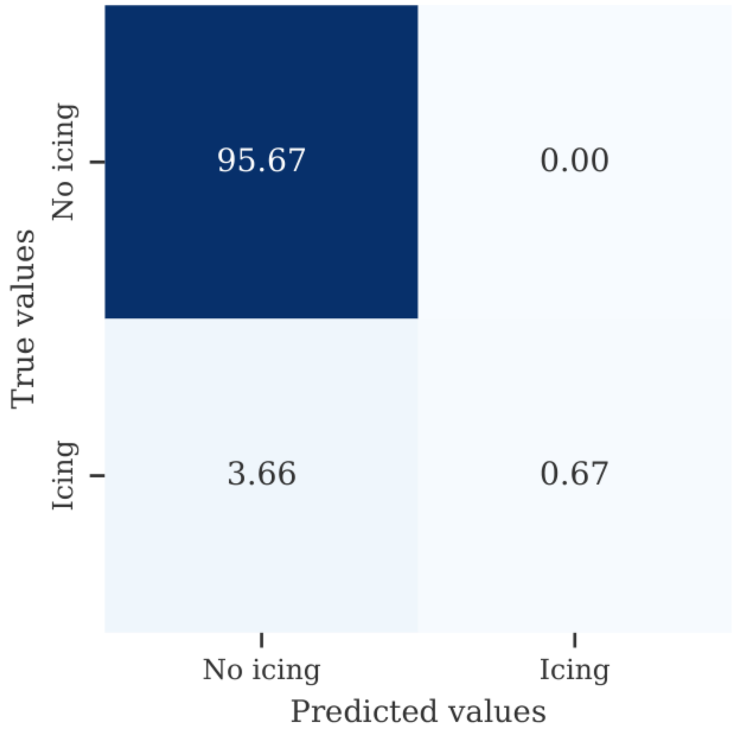
Semester project – CRYOS laboratory

Python

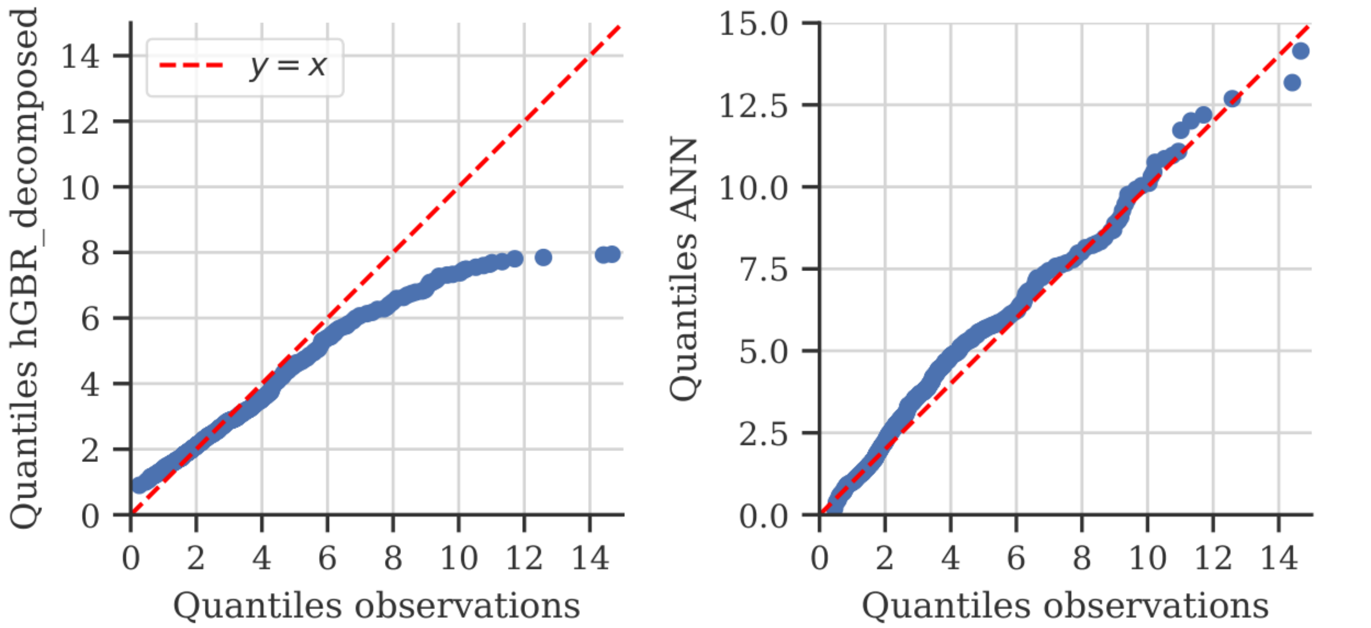
Wind turbine – Data analysis



Icing types 1 & 2

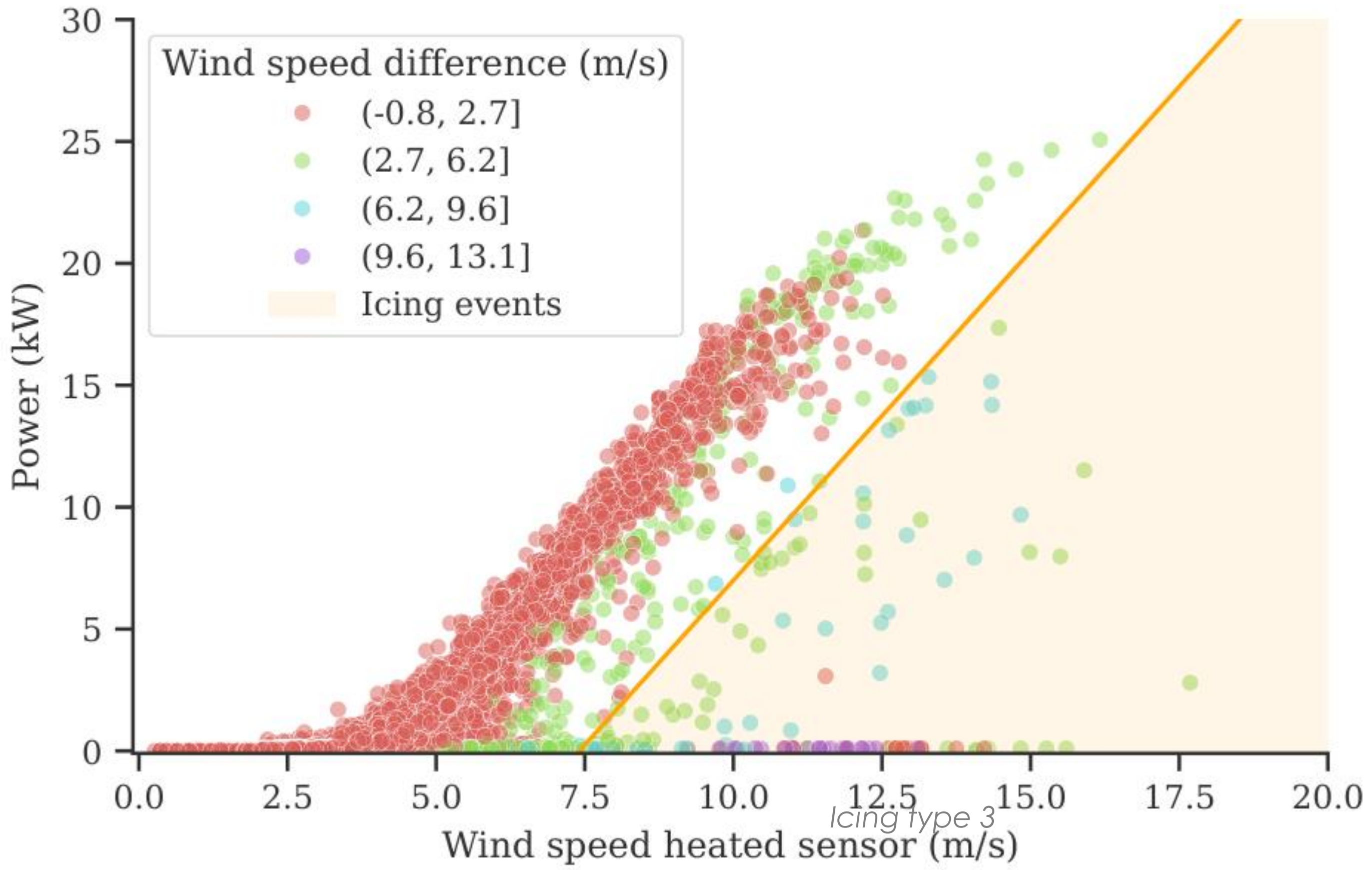


Confusion matrix for icing prediction



(a) q-q plot between observations and hGBR_decomposed

(b) q-q plot between observations and ANN



Icing type 3

Objective Quantify icing-induced production losses and assess the site's long-term wind-energy potential.

Method Labelled icing events using heated and unheated anemometers and trained classification and regression models.

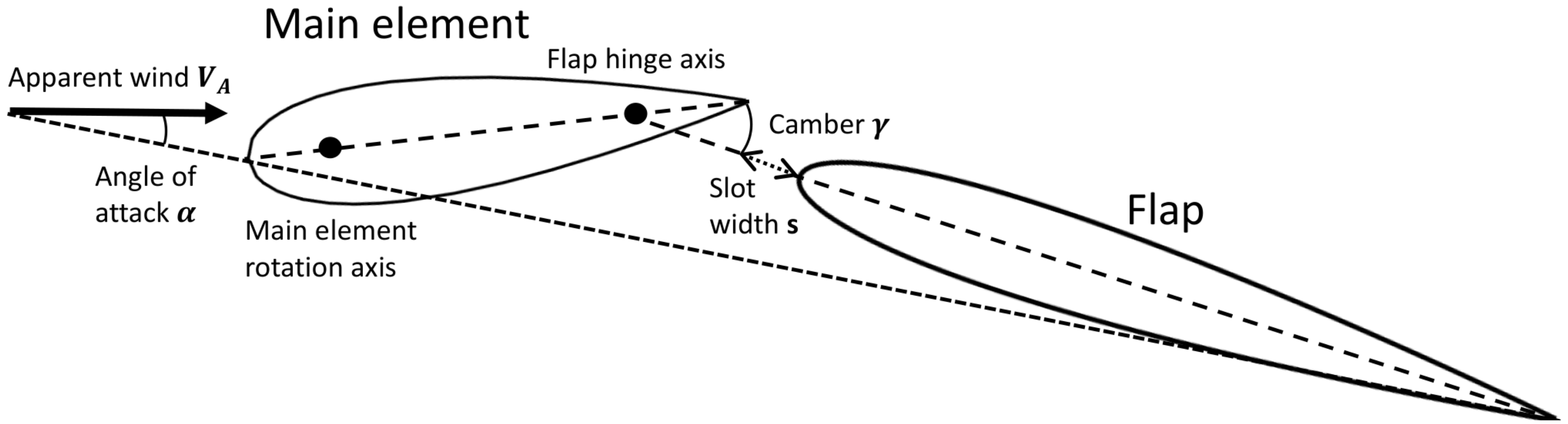
Result Identified 272 icing events, estimated 2.2–7.6 MWh of lost energ (equivalent to 5–18% of production) and achieved 98% icing-classification accuracy.

On the effect of slot width on two-element wingsails

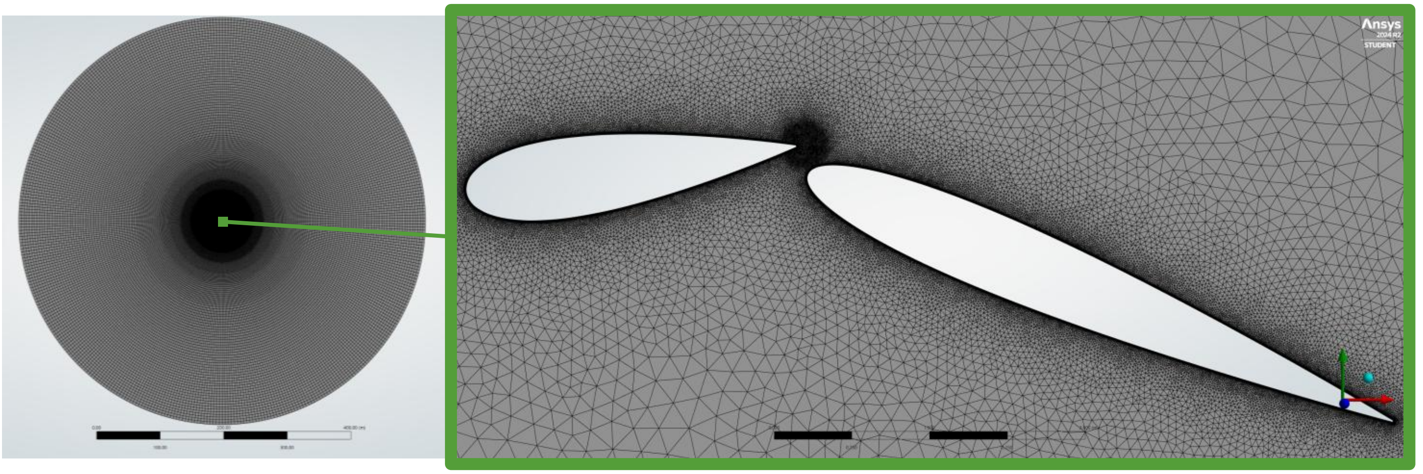
Course project – Numerical Flow Simulation

ANSYS Fluent, Python

Aerodynamics – CFD



Setup



Optimal mesh

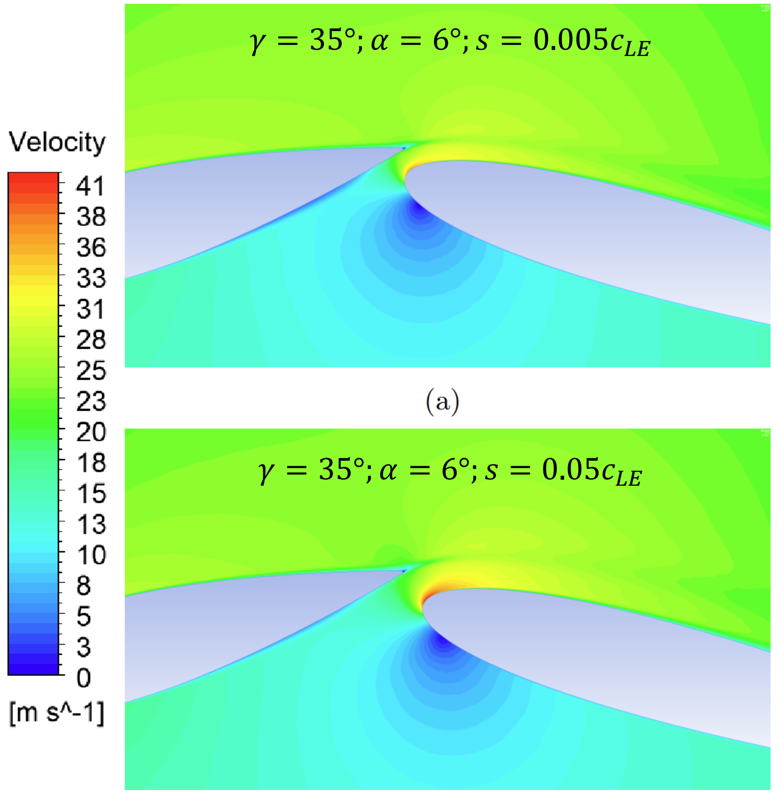
0.5-5% slot width

Re ≈10^7 turbulent regime

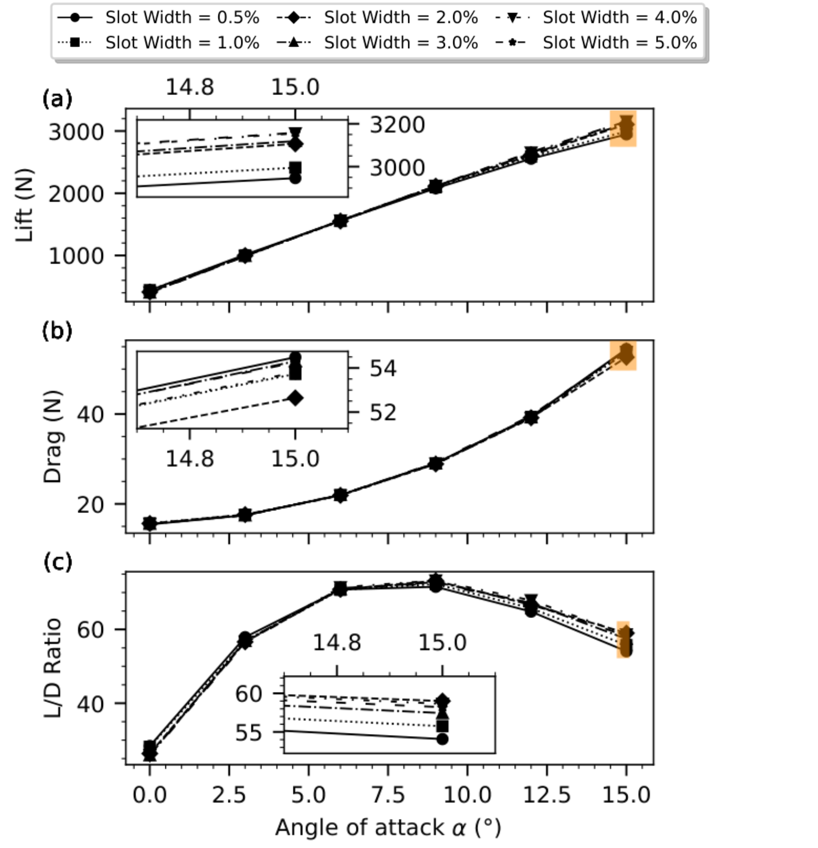
Objective Determine when the slot between two wingsail elements improves aerodynamic performance.

Method Varied angle of attack, camber and slot width after completing mesh- and domain-convergence studies.

Result Slot effectiveness depended strongly on the configuration; wider gaps improved performance at high angles of attack, although stall prediction remained a limitation.



Contour plot of the velocity magnitude



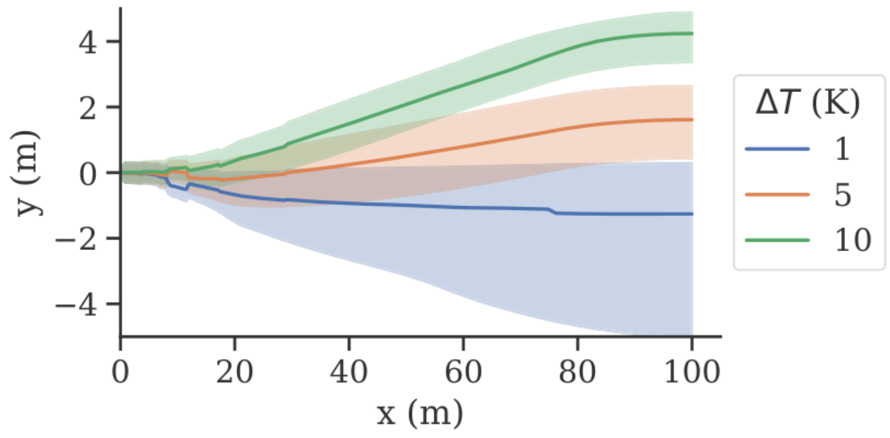
Simulation results showing lift (a), drag (b), and lift-to-drag ratio (c) for various slot widths at a camber angle of 15°

Simulation of particle-laden thermally buoyant horizontal jets

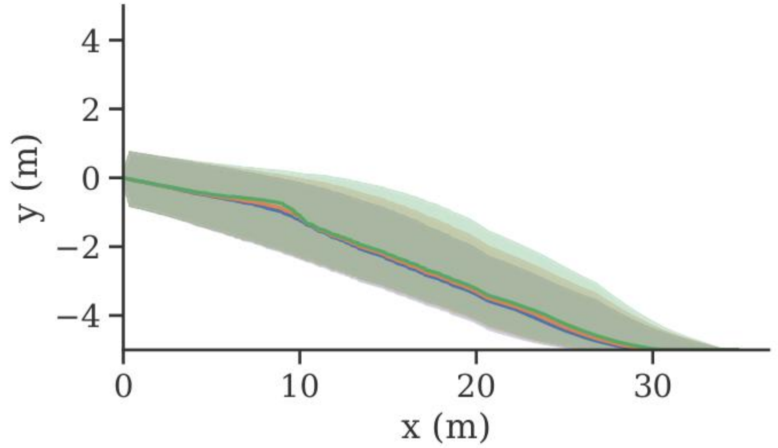
Course project – Environmental Transport Phenomena

ANSYS Fluent

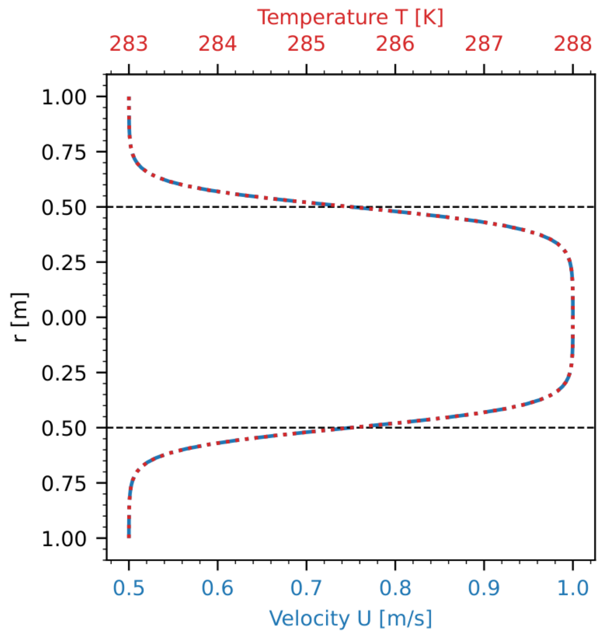
Environmental engineering – CFD



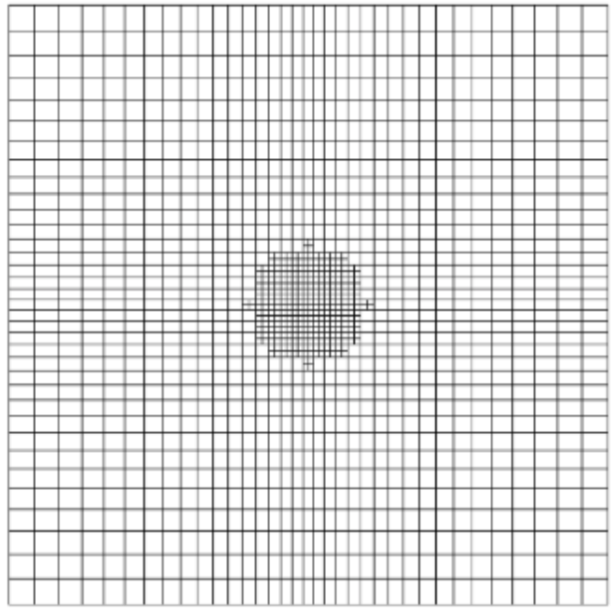
Trajectory of the thermal plume



Trajectory of the sediment plume centerlines



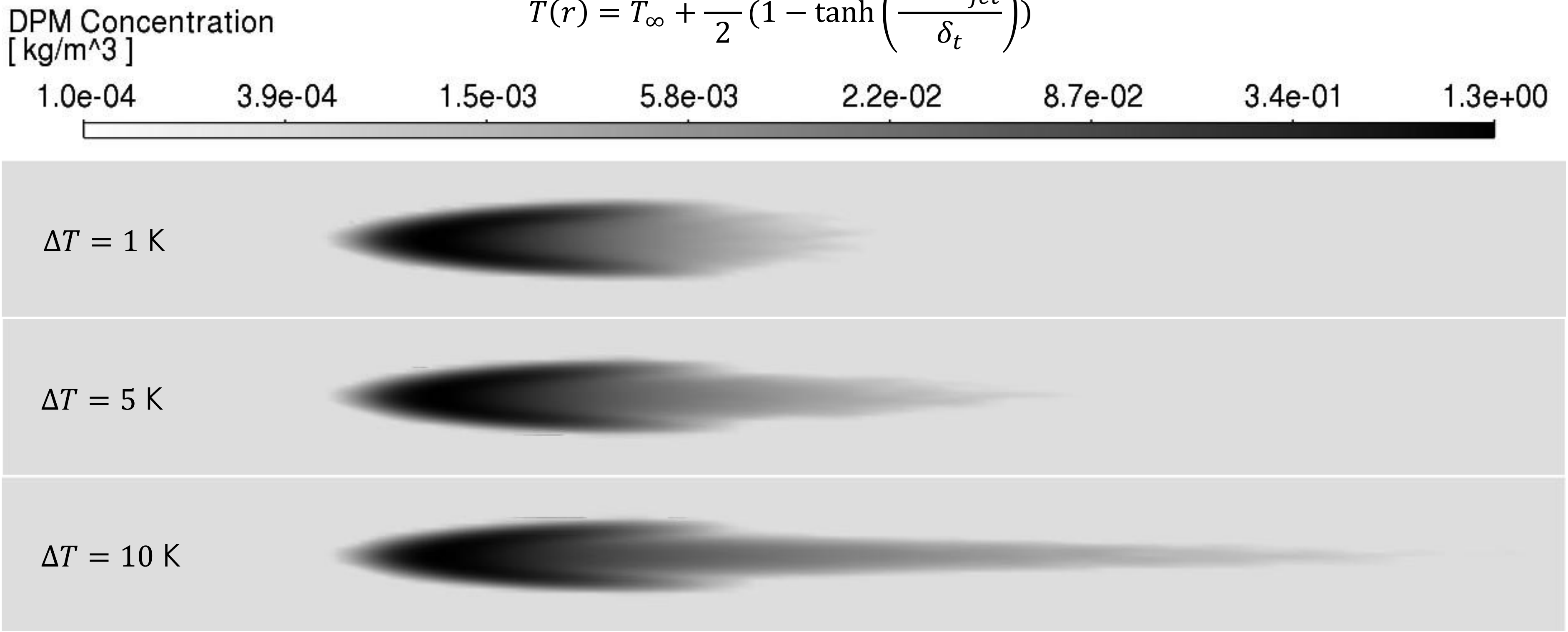
Inlet temperature and velocity profiles



Optimal mesh at the inlet

5x10^4 DPM parcels

$$T(r) = T_\infty + \frac{\Delta T}{2}\left(1 - \tanh\left(\frac{r - R_{jet}}{\delta_t}\right)\right)$$



Sediment deposition maps for different jet temperature

Objective Assess how thermal buoyancy affects jet trajectory and particle deposition.

Method Developed a 3D steady RANS model coupled with a Lagrangian discrete-phase model and verified mesh independence.

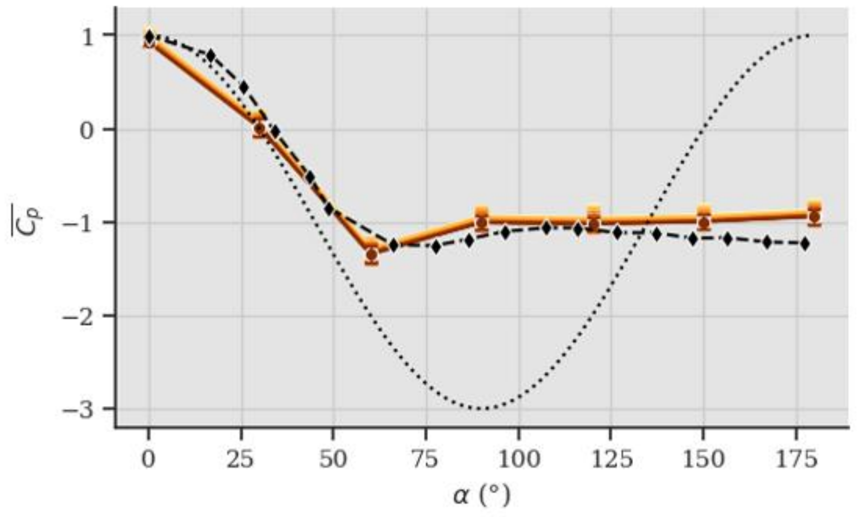
Result Thermal buoyancy strongly deflected the plume, while deposition patterns changed only for a small fraction of particles.

Influence of pneumatic tube length on pressure measurements

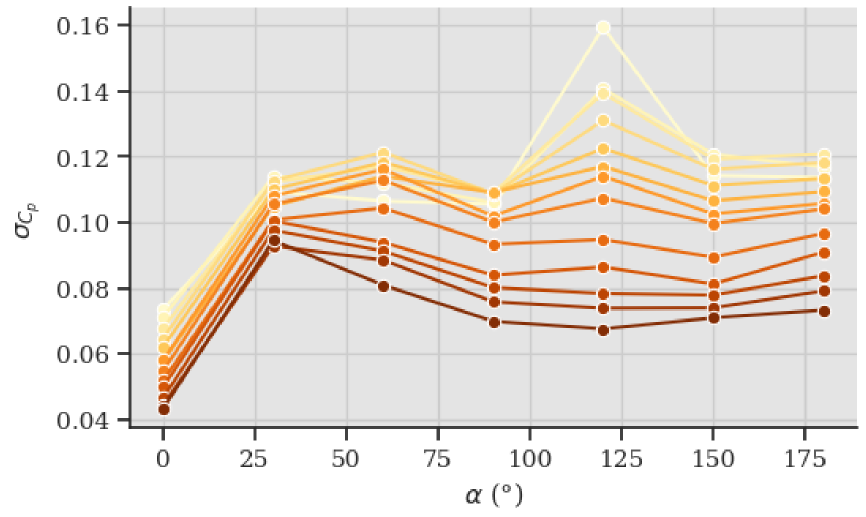
Semester project - UNFoLD laboratory

Python

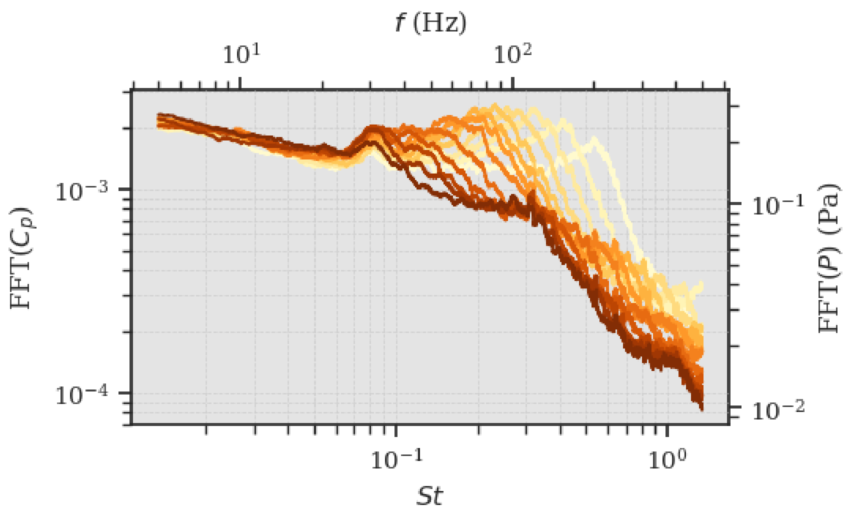
Aerodynamics – Wind tunnel testing



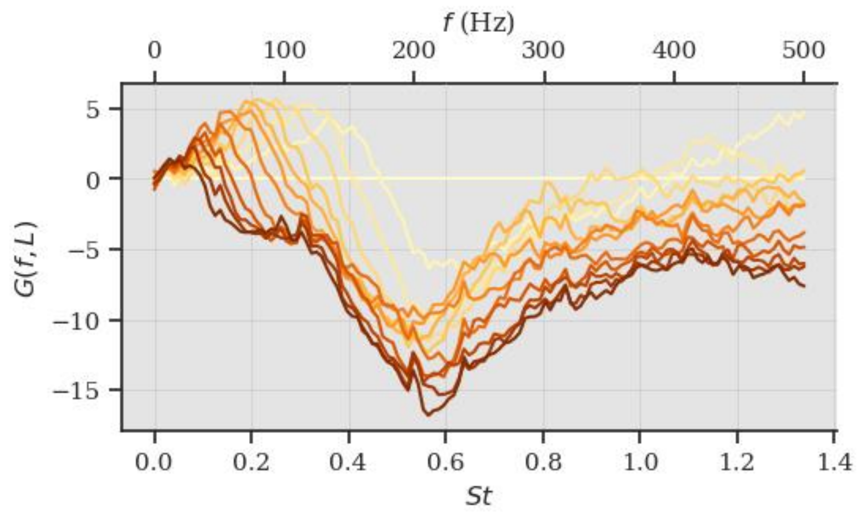
Average pressure coefficient



Standard deviation of pressure coefficient



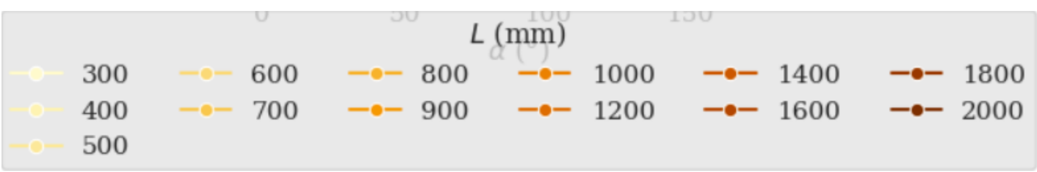
Fourier Transform of the pressure signal for AoA=150°



Spectral gain for AoA=150°

13 tube lengths tested

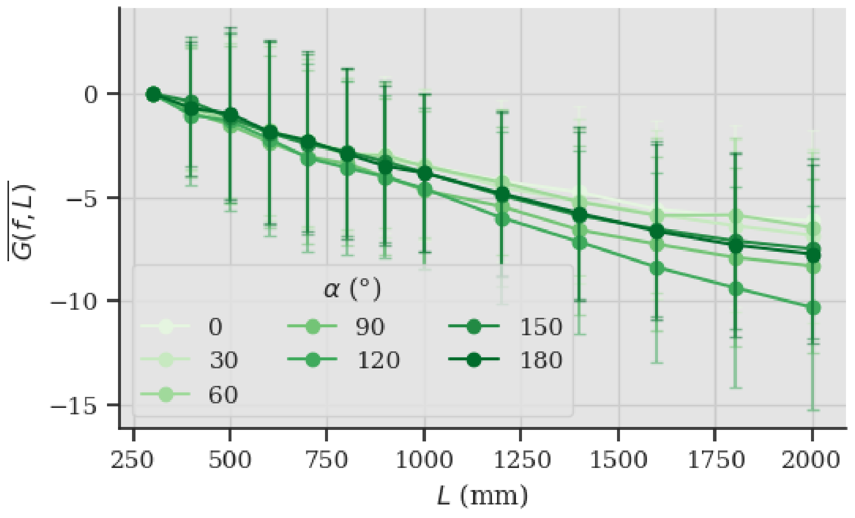
36% fluctuation attenuation



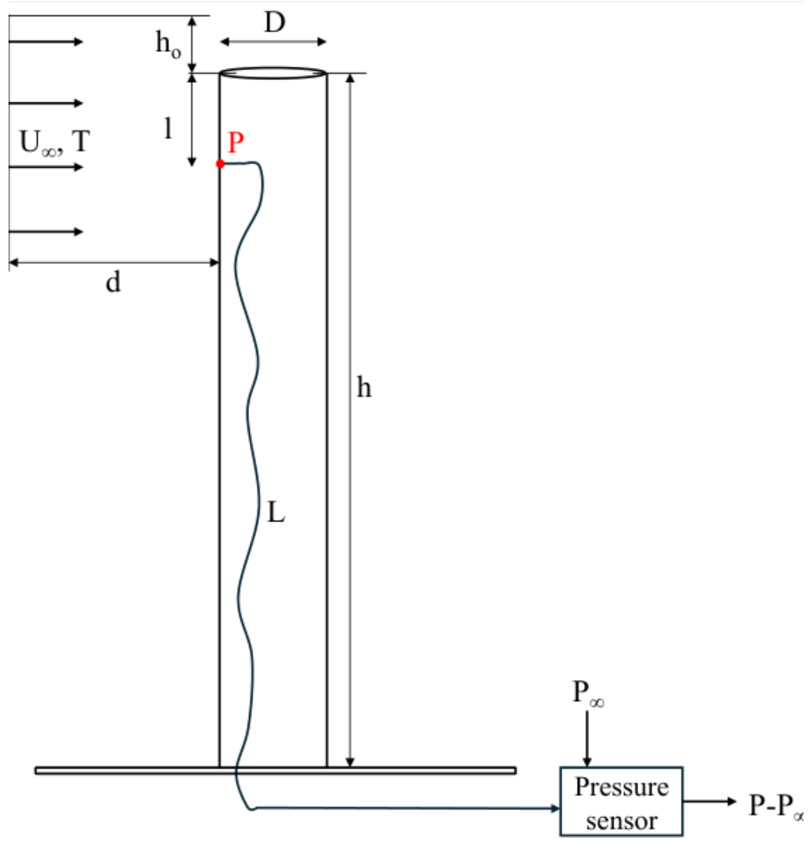
Objective Quantify the effect of pneumatic-tube length on steady and unsteady pressure measurements.

Method Designed the experiment, processed the pressure signals and performed statistical and spectral analyses.

Result Mean pressure coefficients remained nearly unchanged, whereas the longest tube reduced pressure-fluctuation amplitude by approximately 36%.



Average spectral gain for each AoA and tube length



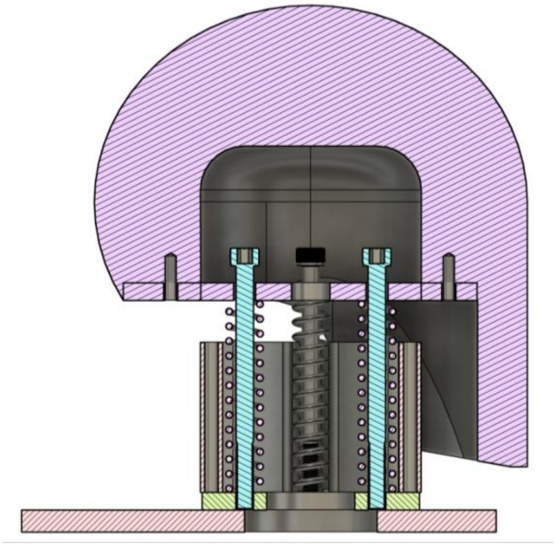
Experimental setup

Improving ski helmet testing using a biofidelic neck model

Course project – Biomechanics of the Musculoskeletal system

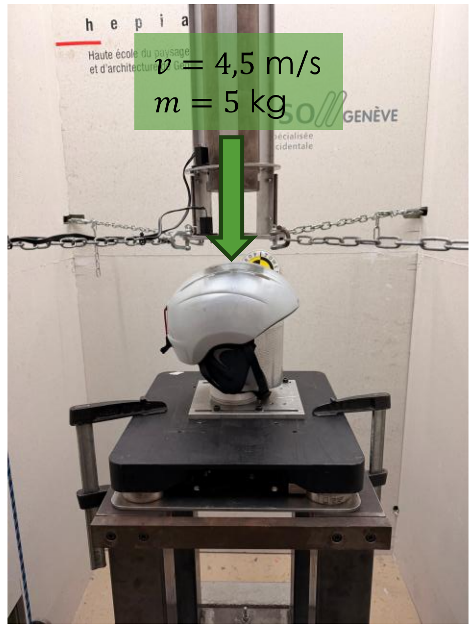
Autodesk fusion, Python

Biomechanics – Crash - CAD



Design of the neck

Photography of the setup



Experimental conditions

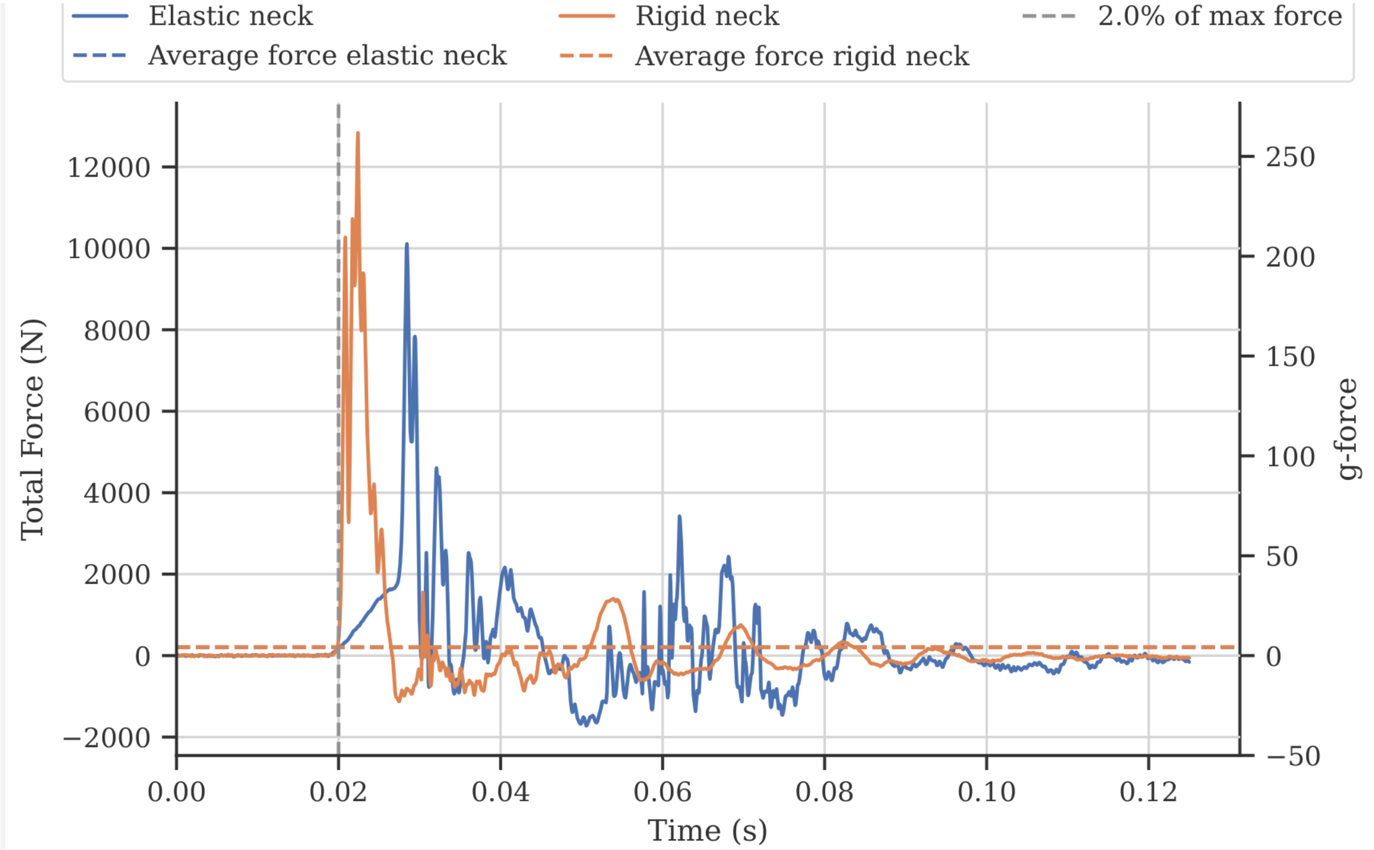
19.2 Hz sampling

12000 N maximum compression

Objective Introduce controlled neck-like compliance into helmet impact tests that conventionally use an isolated rigid head.

Method Designed a manufacturable spring-based neck with guiding elements and an end-stop, then compared rigid and compliant configurations in drop-tower tests.

Result The compliant neck reduced peak impact force from approximately 12–13 kN to 10 kN and reduced the time above 250 g from 52 μs to zero.



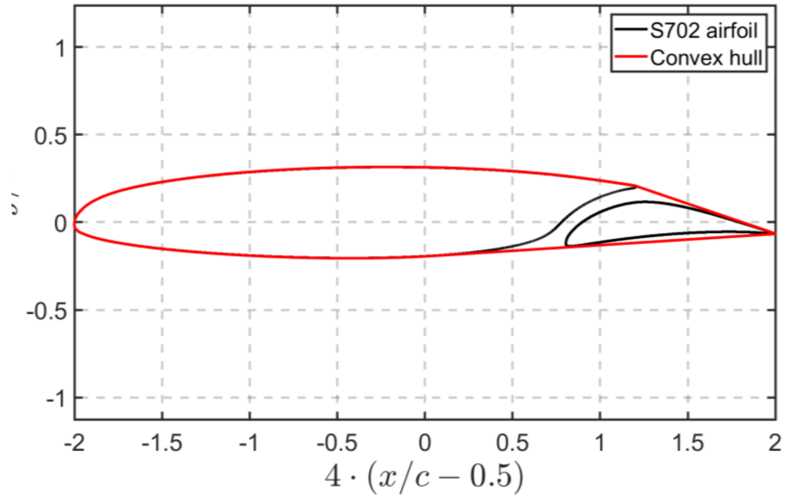
Forces measured by the table with and without the neck

Lift computation of a flapped 2D airfoil

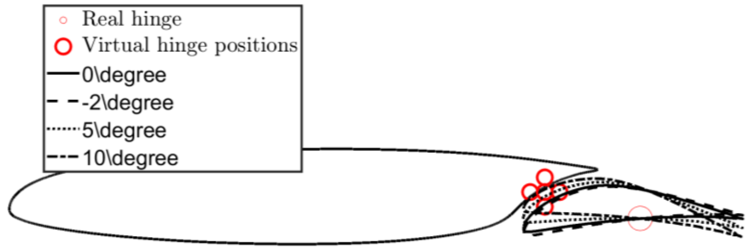
Course project - Aerodynamics

MATLAB

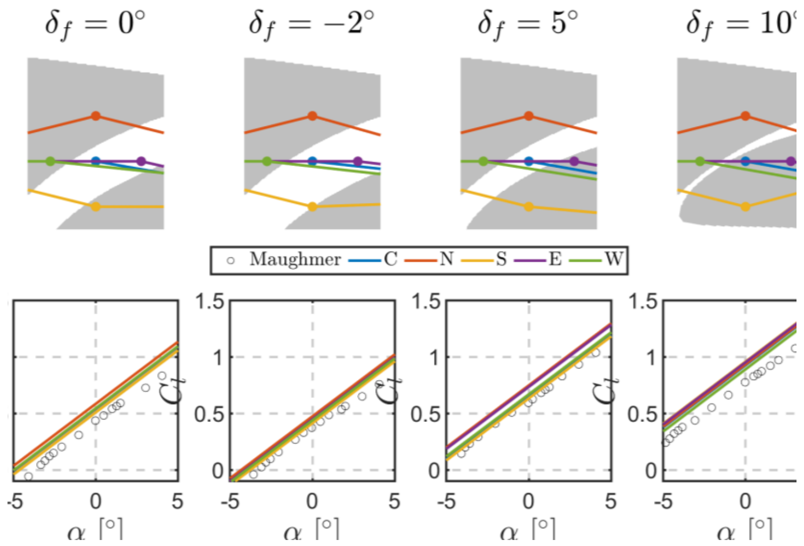
Aerodynamics – Thin Airfoil Theory – Potential flow theory



Convex hull to get the chord line



Defining an arbitrary hinge position



Lift coefficient for each flap deflection and hinge position

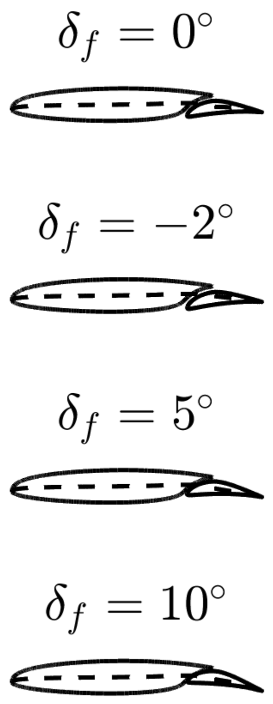
5 different hinge positions

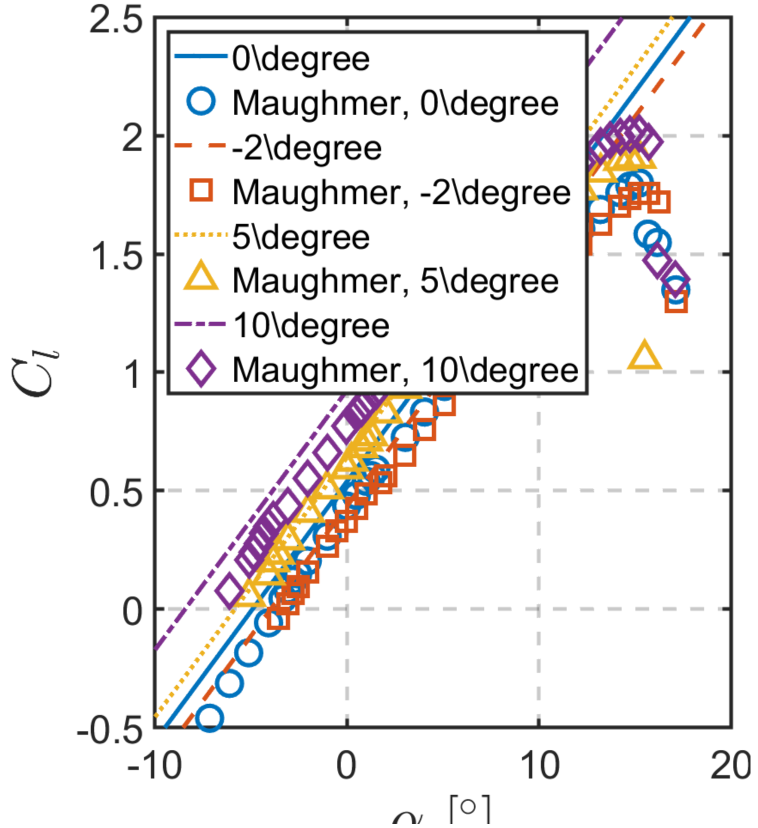
4 flap angles

Objective Reproduce lift and pressure-distribution trends from a published two-dimensional airfoil study.

Method Implemented Thin Airfoil Theory in MATLAB with variable flap deflection and user-defined hinge position, comparing four flap angles and five hinge locations.

Result The model reproduced low-angle attached-flow trends, while deviations increased near separation and stall because inviscid theory does not capture viscous effects.





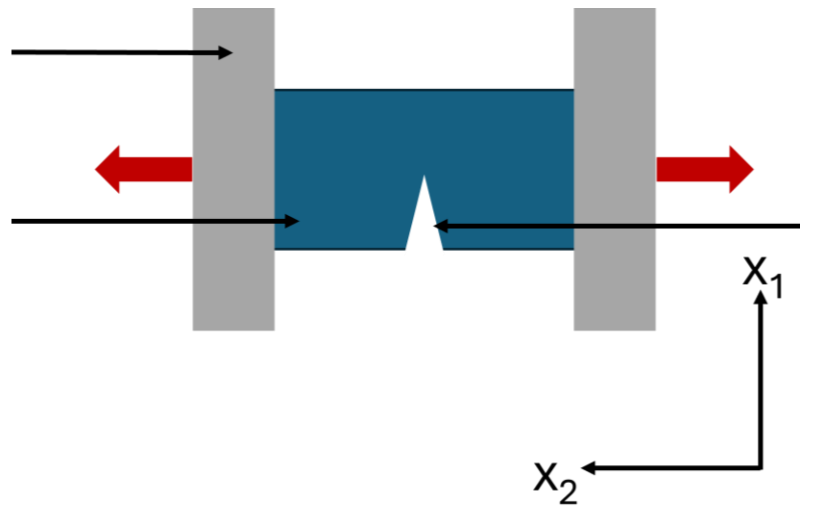
Lift coefficient for each flap deflection and a "center" hinge position

Investigation on the deformation field around a crack tip using Digital Image Correlation (DIC)

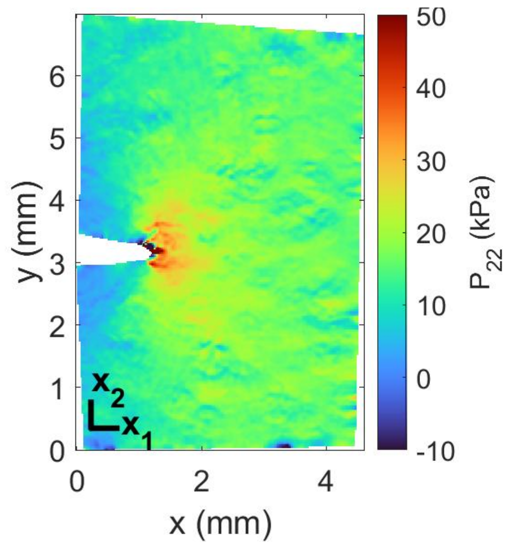
Course Project – Experimental Methods in Mechanical Engineering

Matlab

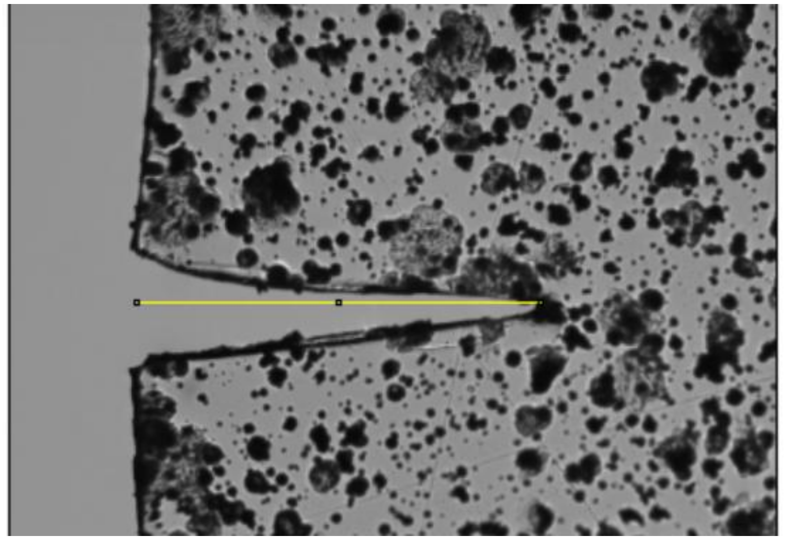
Digital Image Correlation - Structural mechanics



Experimental setup



P_{22} of the Piola-Kirchoff stress tensor



Crack size

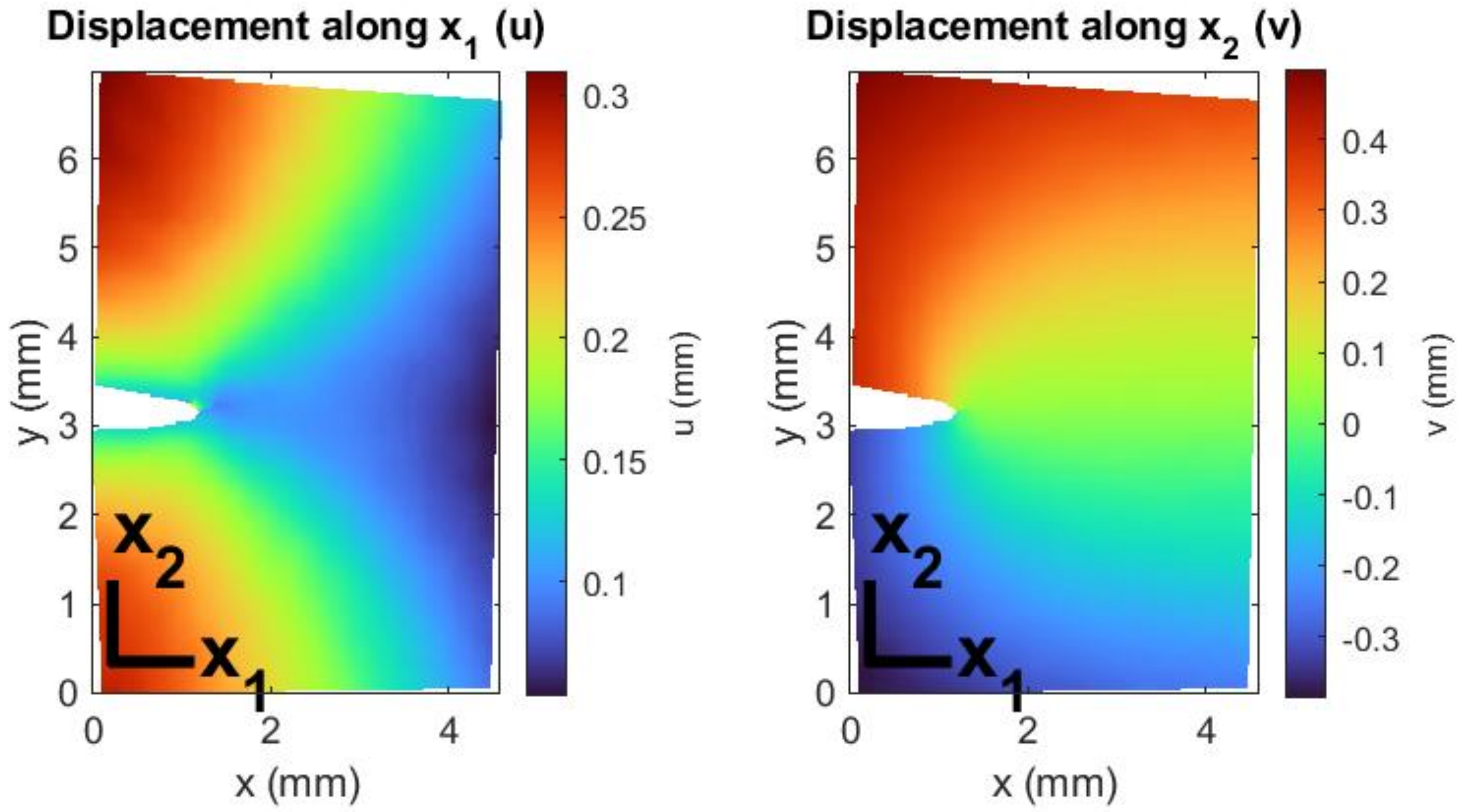
10 frames per second

20 µm/s traction speed

Objective Assess the validity of linear-elastic fracture mechanics for highly deformable polyacrylamide hydrogels.

Method Used Digital Image Correlation to obtain displacement and strain fields, calculated the J-integral and compared linear and nonlinear stress-intensity estimates.

Results The linear model agreed well with two of the three samples, while the third highlighted sensitivity to material nonlinearity and experimental conditions.



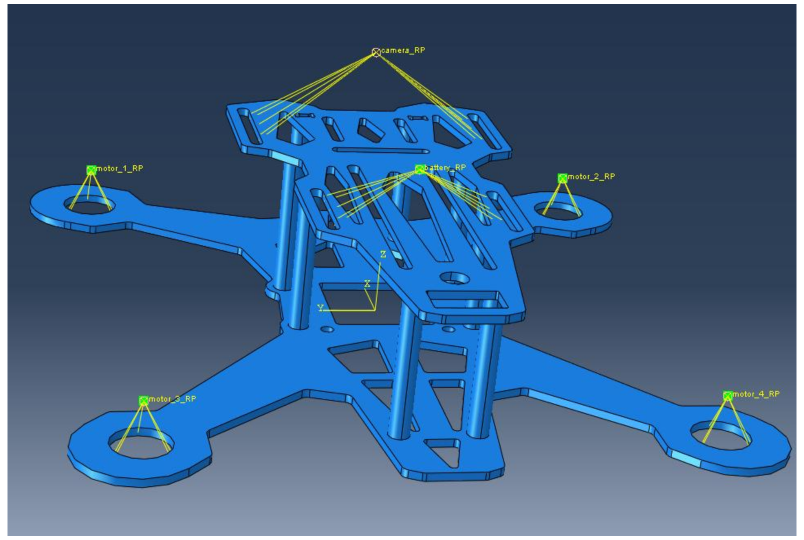
Displacement along each direction

Structural simulation of a quadcopter drone

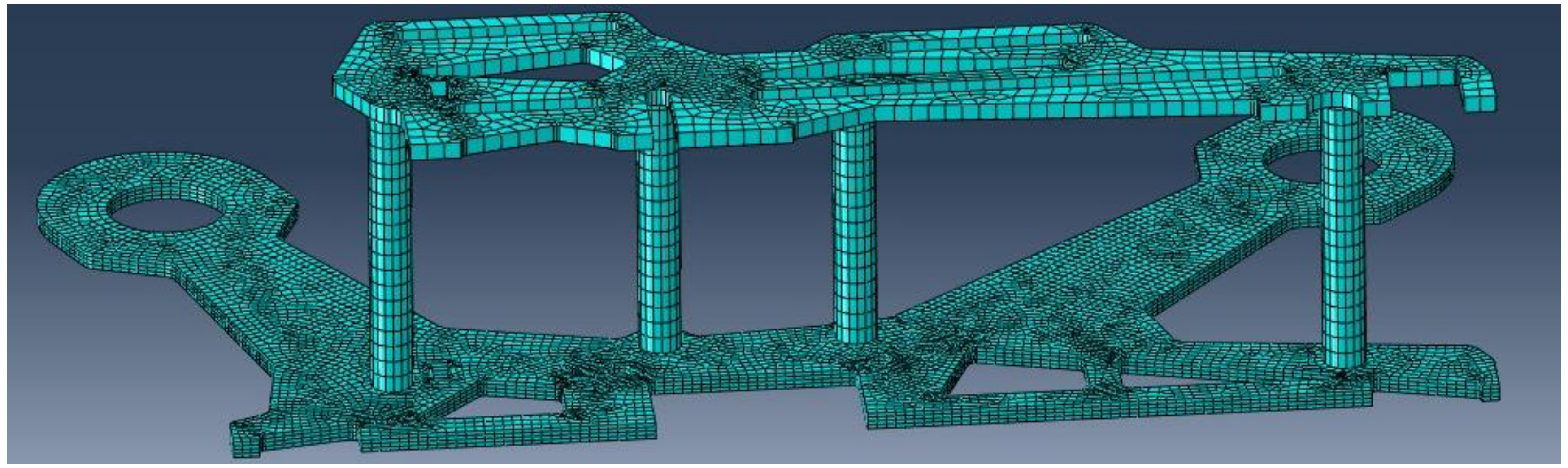
Course project – Méthodes des Elements Finis

Abaqus, Python, Excel

Structural simulation



Simulated drone and loads



Optimal mesh for stationary and camera integration cases

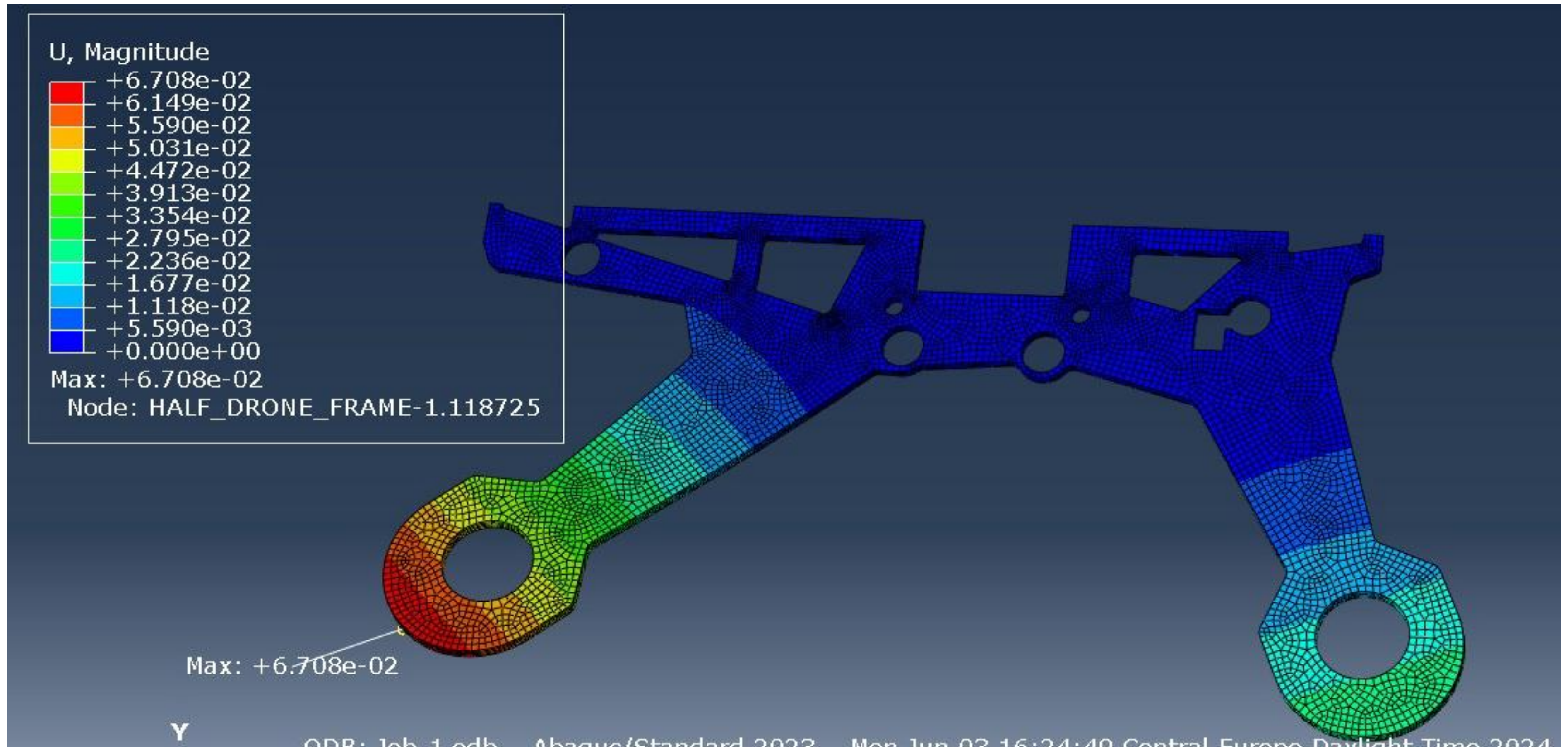
4 load cases

0,17 mm maximum displacement

Objective Assess the structural integrity of an FPV drone frame and identify opportunities for weight reduction.

Method Performed modal and static finite-element analyses with mesh-convergence studies for hovering, camera integration and asymmetric turning conditions.

Results The most demanding turning case produced a maximum von Mises stress of 6.87 MPa and a maximum displacement of 0.17 mm, remaining well below the aluminium yield strength of approximately 259 MPa.



Displacement